Soumil Gupta

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Computer Architecture Researcher experienced in Cycle-Accurate Performance Modeling, Architecture Exploration, RTL & ASIC Design, ML Compilers & AI Accelerator Programming

Seeking New Graduate full-time position in (CPU/GPU/NPU/SoC) Processor Architecture Exploration, Performance Modeling

EDUCATION

Carnegie Mellon University — Expected May 2027

Master of Science in Electrical & Computer Engineering | GPA: 4.0/4.0

University of Illinois, Urbana-Champaign — May 2025

Bachelor of Science in Computer Engineering | GPA: 3.8/4.0 | Honors | Dean's List

Relevant Coursework:

- **Computer Architecture:** Neuromorphic Computer Architecture and Processor Design, Modern Computer Architecture and Systems, Microarchitectural Security, Computer Organization & Design
- **Software:** Operating Systems, Applied Parallel Programming, Communication Networks, Algorithms & Models of Computation, Data Structures & Algorithms, Computer Systems & Programming
- **Hardware:** Advanced Digital IC Design, VLSI System Design, IC Theory & Fabrication, Semiconductor Electronics, Digital Systems, DSP, ASP
- **AI/ML:** Large language Model Systems, Artificial Intelligence, Applied Machine Learning, Computer Vision, IoT & Cognitive Computing

TECHNICAL SKILLS

Languages: C++, C, C#, Python, Java, x86 Assembly

RTL/EDA/Modeling: SystemVerilog, SV Coverage, SystemC, AMD Xilinx Vivado, HLS, Intel Quartus, Synopsys (DC, VCS, Spyglass, Verdi), Verilator, Chipyard, ModelSim, Cadence (Virtuoso, Innovus, Xcelium), UVM, Cocotb, Chisel, PyRTL, Veryl, DynamoRIO, Scarab, Gem5

GPU Systems Tools: CUDA, CSL, Nsight Systems, Nsight Compute, Compute Sanitizer, cuDNN, cuBLAS, TensorRT

ML & Python Scripting Libraries: Tensorflow, Keras, PyTorch, NumPy, SciPy, Scikit-Learn, OpenCV, Pandas, OR-Tools, NetworkX

Tools/Environments: Linux, Git, GDB, Valgrind, Jenkins, Docker, QEMU, GNU Make, LaTeX, Bash, Tcl

WORK EXPERIENCE

SoC Architect Intern, Performance & Power Modeling | Arm — May 2026 - Aug 2026

- Contributed to the 2nd-Gen Arm AGI CPU (AI Datacenter SoC) **architecture specification document (SAR) and customer-facing product specs**; shaping customer performance & power commitments, SKU definitions, die-binning and allocation methodology.
- **Wrote workload performance and power IP-level models in Python** to calculate workload-specific power and performance numbers of candidate SKUs, with die solver to find voltage-frequency operating point for max committed perf. within customer power constraints.
- Built a complete **constrained optimization framework over modeled die population (yield, process variation)** in Python to derive customer volume-aware SKUs, die bin definitions, **globally optimal die-to-die matching, improving projected Perf/TCO by 13.9%.**

CPU Performance Architect Intern | Samsung Research America — Sept 2025 - Dec 2025

- **Modeled leading Arm IP of CPU microarchitecture** components to **assess power, performance, area (PPA) trade-offs,** collaborating with senior engineers at SRA's SoC Architecture Lab for CPU microarchitectural exploration for next-gen Exynos SoCs.
- **Proved IPC increase** of one microarchitectural feature resulting in being added to the team-wide CPU model.
- **Wrote performance models of Arm CPU features using Gem5 and standalone C++ simulation**, emulation, and analysis tools.

Teaching Assistant | Computer Systems & Programming Course, UIUC — Aug 2024 - May 2025

- Led weekly office hours with 5-10 students per session, guiding students in developing algorithms and debugging with GDB.
- **Taught C & C++ programming fundamentals**, and file/device I/O, memory, signals & interrupt handling, and concurrency.

RESEARCH EXPERIENCE

High Performance Computing (HPC) Researcher | SPIRAL Lab, CMU — Jan 2026 - Present

- **Accelerate FFT/NTT, cryptography, ML kernels on the Cerebras WSE-2** by mapping algorithms to the chip's CSL programming model and **profiling and optimizing end-to-end kernel performance.**

Hardware Accelerator Researcher | Future Architecture & Systems Tech. Lab, UIUC — Apr 2024 - May 2025

- **Write, verify, layout, & tapeout proof-of-concept SoC of Berkeley Boom RISC-V Cores, MESA CPU Controller, & CGRA.**
- **Developed a fully custom ISA for CPU controller** to map CPU workloads on any **CGRA Hardware Accelerator via SDFGs.**
- **Built custom hardware verification framework** complete with golden model to compare DUT v. gold execution traces.
- **Verified SystemVerilog, Chisel modules** on AXI units, register files, etc., with ModelSim, Verilator, and Vivado software.

ML Researcher | Autonomous & Unmanned Vehicle Systems Lab (AUVSL), UIUC — Dec 2022 - May 2025

- Led a team to design and implement novel outdoor robot localization solution to solve robot autonomy in GPS-denied areas by leveraging Ultrawide-Band Antennas. **Wrote Python & C++ packages**, deployable on Clearpath robot platforms.
- **Developed PyTorch software** to tune Adaptive Neural Fuzzy Inference System parameters, implemented SHFAF localization algorithm that improved localization accuracy by 82% to within 10 cm. **Wrote Python scripts for data-processing pipeline.**
- Co-authored IEEE paper on localization methods. Presented updates, deliverables to CERL, U.S. Army Corps of Engineers.

SELECTED RESEARCH PUBLICATIONS

1. **S. Gupta**[1], "Implementation of a CPU Controller for Dynamic Binary Translation," May 2025 (Accepted to UIUC IDEALS).
2. M. Juston[1], **S. Gupta**[1], et al., "Robust Error State Sage-Husa Adaptive Kalman Filter for UWB Localization," IEEE Sensors Journal, vol. 25, no. 9, pp. 16034–16049, Mar 2025, doi: 10.1109/JSEN.2025.3549315.

***3 publications in computer architecture from my M.S. program currently in progress (1. Neuromorphic Computing-Based GEMM Unit for ML Model Training, 2. Reconfigurable Cryptography Accelerator, 3. HPC Kernels for Arithmetically-Intense Computations)**

APPLE-SPONSORED ASIC TAPEOUT

CAESAr – Crypto-Agile Encryption Streaming Accelerator — May 2026

- Designed a reconfigurable cryptographic accelerator to offload encryption and authentication security operations from the CPU, with the purpose of freeing processor cycles while supporting current and emerging AES modes without need to redesign hardware.
- Successfully Taped-out a complete SoC around CAESAr accelerator with a VexRisc RV32IMC CPU, 128-bit AXI4 fabric, 32 KB DMA-fed SRAM, JTAG/UART interfaces, and separate 400 MHz CPU and 1 GHz accelerator clock domains.
- Implemented pipelined AES-128/256 and GF(2^{128}) GHASH/POLYVAL datapaths supporting GCM, GCM-SIV, HCTR2, XTS, EME2, ECB, and CTR at one 128-bit block per cycle; Performed RTL verification using with NIST and RFC 8452 vectors.
- Implemented own RISC-V crypto-extension and software tests to benchmark against accelerator, proving benefit of CAESAr solution.
- Closed physical design and signoff in TSMC 28nm HPC+ using Xcelium, Genus, Innovus, Voltus, and Calibre with Tcl; Resulting in 1.2 mm x 1.0 mm ASIC tapeout with 135 mW average post-layout power.
  *Post-silicon validation in progress, currently preparing a conference paper to be published on CAESAr.

SELECTED HARDWARE SYSTEMS PROJECTS

STEWARD – CPU Architectural Feature for Thread Context Switches in Scarab Simulator — May 2026

- Designed STEWARD, a hardware-managed state-restoration mechanism that rebuilds a thread's L1 instruction cache, branch target buffer (BTB), branch predictor, and return-address stack to cut datacenter context-switch warm-up penalties in serverless systems.
- Built cycle-accurate CPU models in C++ in Scarab, a trace-driven out-of-order x86 simulator; extended DynamoRIO to inject context switches into Google datacenter traces, flush in-flight work, and save/restore per-thread microarchitectural state.
- Implemented adaptive cache-only or cache-plus-BTB restoration with bounded continuous BTB replay and deduplicated L1I prefetching, improving IPC 65% over baseline and 28% over Ignite, the prior state-of-the-art restoration mechanism.

RISC-V 32IM N-Way Superscalar Out-of-Order Explicit Register Renaming CPU in SystemVerilog — May 2024

- Designed & verified from scratch an out-of-order N-way parameterizable processor for RISC-V 32-bit ISA with M-extension, achieving IPC of 0.51 on standard benchmark taking 21.7 mW of power at clock frequency 325 MHz.
- Implemented speculative branching with branch prediction overriding, & perceptron and Gshare branch predictors.
- Implemented cache features including next-line & stride prefetchers, post-commit store buffer with write coalescing.
- Integrated Synopsys IPs including sequential divider, and wrote Dadda advanced multiplier and shift-add multiplier.
- Wrote full processor in SystemVerilog, Simulated with Synopsys VCS, Debugged & Verified with Synopsys DC, Verdi, Spike, RVFI, Lint, Functional Coverage, Used Python to generate test programs, Built toolchains with bash, GNU make.

RNS-tubGEMM – Residue Number System-Based General Matrix-Multiply (GEMM) Unit — May 2026

- Designed RNS-tubGEMM, a matrix-multiply (GEMM) accelerator to reduce matrix-multiply main area and energy cost.
- Implemented parameterized 4x4, 8x8, and 16x16 RNS processing-element arrays in SystemVerilog with input conversion, operand streaming, parallel multiply-accumulate datapaths; picked format representation converter via PPA exploration.
- Built an analytical model of latency, idle time, array utilization, and zero-work steps to assess resource utilization and wasted work.
- Synthesized the accelerator at 400 MHz in NanGate 45nm using Cadence Genus, evaluated post-synthesis PPA scaling, and proved 43.0x and 15.7x temporal-latency improvements over baseline temporal-unary GEMM unit.

SELECTED AI/ML/SOFTWARE SYSTEMS PROJECTS

OpenAI GPT-2 Transformer Full Implementation on NVIDIA H100 GPU — May 2025

- Built a GPU-native MiniTorch deep-learning framework from the CUDA tensor backend through automatic differentiation, then used it to implement, train, and run an OpenAI GPT-2 decoder end-to-end on NVIDIA H100 GPUs.
- Wrote strided/broadcast CUDA map, zip, reduce, and tiled GEMM kernels with shared-memory tiling and Python backend integration. Implemented reverse-mode autodiff/backprop plus Linear, Dropout, LayerNorm, Embedding, softmax loss, and end-to-end training.
- Built GPT-2 masked multi-head attention, pre-LN transformer blocks, embeddings and generation; trained on IWSLT14 dataset, then fused attention Softmax/LayerNorm forward+backward in CUDA with CUB, warp shuffles and float4 vectorization.

FPGA-Accelerated DNN for Autonomous Traffic Sign Recognition — Apr 2025

- Designed a hardware-accelerated Deep Neural Network for AMD PYNQ-Z2 FPGA using AMD Xilinx Vivado HLS.
- Implemented convolution, activation, and pooling layers with HLS pragmas to optimize dataflow and pipelining.
- Leveraged ScaleHLS framework & hardware-software co-design, balancing accuracy with FPGA resource limits.
- Employed TensorFlow to train & deploy models on GPUs, Raspberry Pi, Google Edge TPU for benchmarking.

Custom UNIX Operating System — Dec 2023

- Wrote custom UNIX OS for emulated single-core Intel Pentium IV Processor from scratch using C & x86 Assembly with a paging-only virtual memory system, writable file system, context switching, terminal switching, and hardware interrupts and exception handling.
- Developed device drivers for keyboard, mouse, real-time clock, & built interactive shell for executing system calls.
- Used GDB to efficiently debug and resolve system-level issues. Emulated and tested the OS in a QEMU virtual environment.